

Practical Task: Employee Management REST API

Build a **Employee Management REST API** using **ASP.NET Core Web API** using **Entity Framework Core**.

1. Task Objective

Develop a REST API to manage employees of an organization.

The API must support:

- Employee Model creation
- SQL Server database integration
- Entity Framework Core Code First
- Migration and database creation
- DTOs for API requests/responses
- CRUD API endpoints
- FluentValidation with defined validation rules
- Common API response format
- Proper HTTP status codes
- Error handling through the common response

2. Employee Information

Field	C# Data Type	SQL Server Data Type	Description
EmployeeId	int	int	Primary Key
EmployeeName	string	varchar(100)	Employee full name
EmailAddress	string	varchar(100)	Employee email address
MobileNumber	string	varchar(100)	Employee mobile number
EmployeeCode	string	varchar(100)	Employee code, optional
Salary	decimal	decimal(18,2)	Employee salary
JoiningDate	DateTime	datetime	Date employee joined
BirthDate	DateTime	datetime	Employee birth date
IsActive	bool	bit	Active/inactive status

3. API Endpoints

Implement the following endpoints.

HTTP Method	Endpoint	Purpose	Type
POST	/api/employees	Add a new employee	Create
GET	/api/employees	Get all employees	Read
GET	/api/employees/{id}	Get employee by ID	Read
PUT	/api/employees/{id}	Update employee	Update
DELETE	/api/employees/{id}	Delete employee	Delete
GET	/api/employees/active	Get only active employees	Filtering
GET	/api/employees/search?name=rahul	Search employees by name	Query Parameter
GET	/api/employees/salary-range?min=30000&max=60000	Get employees within salary range	Multiple Query Parameters
GET	/api/employees/joined-between?from=2025-01-01&to=2025-12-31	Get employees who joined within a date range	Date Filtering
GET	/api/employees/summary	Return total, active, inactive employees and average salary	Aggregation
GET	/api/employees/highest-paid	Return the employee with the highest salary	Sorting/Aggregation

4. FluentValidation Requirements

Implement FluentValidation for the Employee.

#	Field	Validation Requirement
1	EmployeeName	Required
2	EmployeeName	Must not contain leading/trailing whitespace
3	EmployeeName	Minimum 3, Maximum 100 characters characters
4	EmployeeName	Must contain only valid name characters
5	EmailAddress	Valid Email Address, No Whitespace
6	MobileNumber	Required, Must contain only digits
7	Salary	Must be greater than <code>0</code>
8	Salary	Must not exceed <code>10,00,000</code>
9	JoiningDate	Required
10	JoiningDate	Must not be a future date
11	IsActive	Must contain a valid boolean value

5. Common API Response

All API endpoints should return a **common response structure** instead of returning raw objects directly.

The response should conceptually contain:

Property	Purpose
<code>Status</code>	Indicates whether the operation was successful
<code>Message</code>	Human-readable operation/result message
<code>Data</code>	Actual response data; can be <code>null</code>
<code>Error</code>	Error information; can be <code>null</code>

6. HTTP Status Code Requirements

The intern should use appropriate HTTP status codes.

Scenario	Expected Status
Successful GET	200 OK
Successful POST	201 Created
Successful PUT	200 OK
Successful DELETE	200 OK or 204 No Content
Validation Failure	400 Bad Request
Employee Not Found	404 Not Found
Unexpected Server Error	500 Internal Server Error
